MOHAN BABU UNIVERSITY

Sample Academic Assignment

Subject: IoT

Student Name: _____
Student ID: _____
Submission Date: _____

Objective

Design a basic Internet of Things solution for a smart classroom.

Scenario

A classroom wants to monitor temperature, humidity, occupancy, and energy usage.

Tasks

1. Draw a system architecture diagram showing sensors, microcontroller, network, cloud/server, and user dashboard.

2. Choose suitable sensors and a controller (for example, ESP32) and justify your choices.

3. Describe how data travels from the sensors to the dashboard.

4. Suggest one communication protocol and explain why it fits.

5. Write pseudocode for collecting readings and sending them to a server.

6. Identify three security or privacy risks and propose mitigations.

7. Describe two useful alerts or automated actions for the classroom.

Submission

Submit the architecture diagram, written explanation, pseudocode, and a short conclusion.

Note: This is a sample practice assignment created for website testing and learning. It is not an official university-issued assignment.